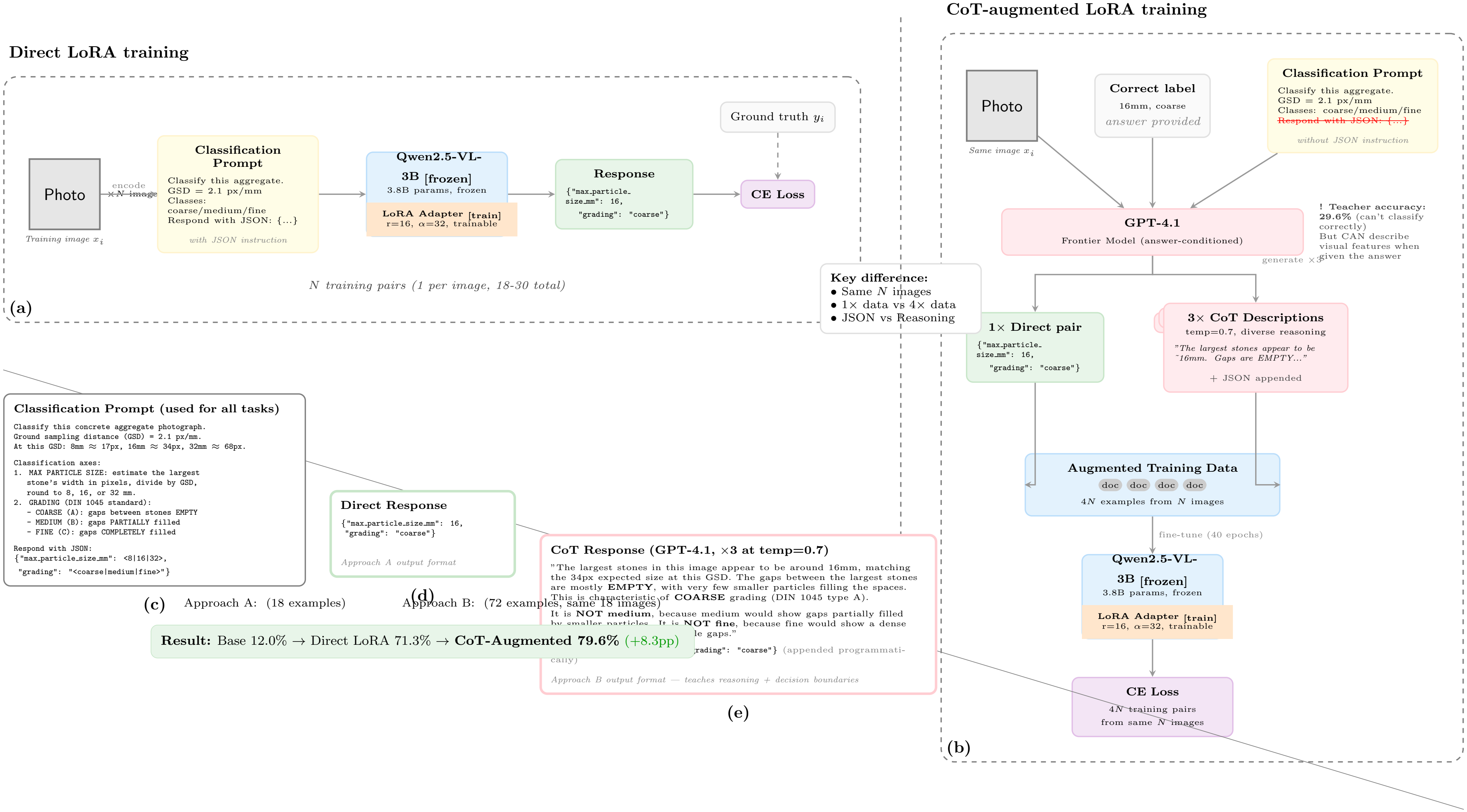
Direct LoRA training
Photo
Training image x_i
encode ×N images
Classification Prompt
Classify this aggregate.
GSD = 2.1 px/mm
Classes: coarse/medium/fine
Respond with JSON: {...}
with JSON instruction
Qwen2.5-VL-3B [frozen]
3.8B params, frozen
LoRA Adapter [train]
r=16, α=32, trainable
Response
{"max_particle_size_mm": 16, "grading": "coarse"}
Ground truth y_i
CE Loss
N training pairs (1 per image, 18-30 total)
(a)
Key difference:
• Same N images
• 1× data vs 4× data
• JSON vs Reasoning
CoT-augmented LoRA training
Photo
Same image x_i
Correct label
16mm, coarse
answer provided
Classification Prompt
Classify this aggregate.
GSD = 2.1 px/mm
Classes: coarse/medium/fine
~~Respond with JSON: {...}~~
without JSON instruction
GPT-4.1
Frontier Model (answer-conditioned)
! Teacher accuracy: 29.6% (can't classify correctly)
But CAN describe visual features when given the answer
generate ×3
1× Direct pair
{"max_particle_size_mm": 16, "grading": "coarse"}
3× CoT Descriptions
temp=0.7, diverse reasoning
"The largest stones appear to be ~16mm. Gaps are EMPTY..."
+ JSON appended
Augmented Training Data
doc doc doc doc
4N examples from N images
fine-tune (40 epochs)
Qwen2.5-VL-3B [frozen]
3.8B params, frozen
LoRA Adapter [train]
r=16, α=32, trainable
CE Loss
4N training pairs from same N images
(b)
Classification Prompt (used for all tasks)
Classify this concrete aggregate photograph.
Ground sampling distance (GSD) = 2.1 px/mm.
At this GSD: 8mm ≈ 17px, 16mm ≈ 34px, 32mm ≈ 68px.
Classification axes:
1. MAX PARTICLE SIZE: estimate the largest stone's width in pixels, divide by GSD, round to 8, 16, or 32 mm.
2. GRADING (DIN 1045 standard):
- COARSE (A): gaps between stones EMPTY
- MEDIUM (B): gaps PARTIALLY filled
- FINE (C): gaps COMPLETELY filled
Respond with JSON:
{"max_particle_size_mm": <8|16|32>, "grading": "<coarse|medium|fine>"}
(c) Approach A: (18 examples)
Direct Response
{"max_particle_size_mm": 16, "grading": "coarse"}
Approach A output format
(d) Approach B: (72 examples, same 18 images)
CoT Response (GPT-4.1, ×3 at temp=0.7)
"The largest stones in this image appear to be around 16mm, matching the 34px expected size at this GSD. The gaps between the largest stones are mostly EMPTY, with very few smaller particles filling the spaces. This is characteristic of COARSE grading (DIN 1045 type A).
It is NOT medium, because medium would show gaps partially filled by smaller particles. It is NOT fine, because fine would show a dense ... le gaps."
... rading": "coarse"} (appended programmatically)
Approach B output format — teaches reasoning + decision boundaries
(e)
Result: Base 12.0% → Direct LoRA 71.3% → CoT-Augmented 79.6% (+8.3pp)